

Detecting Physiological Responses Using Multimodal Earbud Sensors

Md Mahbubur Rahman^{1*}, Xuhai Xu^{2*}, Viswam Nathan¹, Tousif Ahmed¹, Mohsin Yusuf Ahmed¹,
Dan McCaffrey¹, Jilong Kuang¹, Trevor Cowell³, Julia Moore³, Wendy Berry Mendes³ and Jun Alex Gao¹

Abstract—Continuous stress exposure negatively impacts mental and physical well-being. Physiological arousal due to stress affects heartbeat frequency, changes breathing pattern and peripheral temperature, among several other bodily responses. Traditionally stress detection is performed by collecting signals such as electrocardiogram (ECG), respiration, and skin conductance response using uncomfortable sensors such as a chestband. In this study, we use earbuds that passively measure photoplethysmography (PPG), core body temperature, and inertial measurements. We have conducted a lab study exposing 18 participants to an evaluated speech task and additional tasks aimed at increasing stress or promoting relaxation. We simultaneously collected PPG, ECG, impedance cardiography (ICG), and blood pressure using laboratory grade equipment as reference measurements. We show that the earbud PPG sensor can reliably capture heart rate and heart rate variability. We further show that earbud signals can be used to classify the physiological responses associated with stress with 91.30% recall, 80.52% precision, and 85.12% F1-score using a random forest classifier with leave-one-subject-out cross-validation. The accuracy can further be improved through multi-modal sensing. These findings demonstrate the feasibility of using earbuds for passively monitoring users' physiological responses.

I. INTRODUCTION

Various stressors in daily life can trigger physiological responses that may either be detrimental to mental and physical well-being, or be useful to increase productivity [20]. However, repeated exposure to stress can accumulate and is linked to cardiovascular diseases and essential hypertension [1], [2], [3], [4]. Continuous physiological stress measurement during experiences of stress combined with negative stress exposure from environmental context can help generate rich insights for stress management.

Stress monitoring approaches focus on physiological system activation, which responds to physical and psychological demands. The two main pathways by which stress affects the body are the hypothalamic-pituitary-adrenal (HPA) axis and the sympathetic adrenal medullary system (SAM) [25]. The HPA axis triggers the adrenal cortex which results in cortisol release [25]. Assessment of cortisol in body fluid such as blood, saliva, or urine is considered the gold-standard stress measurement. However, such assessment is less suitable for

continuous, passive stress monitoring due to the complexities in sampling.

Whereas, SAM activation can be measured by monitoring the autonomic nervous system (ANS). The ANS quickly promotes physiological changes through the sympathetic nervous system (SNS) and the parasympathetic nervous system (PNS). The SNS is associated with energy mobilization and the PNS is associated with restorative function [26]. Relative increase in the SNS activity leads to increases in heart rate. Furthermore, relative increases in the PNS activity leads to decrease in heart rate, although the two subsystems are not always reciprocal. Therefore, the analysis of time intervals between heartbeats, referred as heart rate variability (HRV) analysis, provides information on relative activation of the SNS and PNS.

HRV analysis requires information regarding the time between each heartbeat known as inter-beat intervals (IBIs). Thus, common heart rate monitors, which average heart rate over tens of seconds, are not sufficient. Previous studies [3] using ambulatory monitoring devices have the potential to monitor HRV data for continuous stress in daily life, however monitors using chest straps or ECG sensors can be uncomfortable and interfere with daily activities. Recent works show that earbuds can reliably capture PPG data and estimate inter-beat-intervals [8]. However, the PPG sensor by itself may not be enough for stress detection models that examine ANS cardiac control across subjects. This is mainly because heart rate variability can also be affected by respiration rate, tidal volume (displaced air during respiration) and posture (supine/standing) [24].

We have conducted a feasibility study with 18 healthy subjects, where we have collected raw PPG, 3-axis accelerometer signals, and core body temperature (CBT) using earbuds. Moreover, we have collected reference physiological sensor data using laboratory-grade research equipment (Biopac, Goleta, CA) [9]. The reference equipment includes electrocardiography (ECG), impedance cardiography, ear plethysmography, and a cuffed blood pressure monitor (Space Labs). During the study participants have completed an evaluated speech task, similar to the Trier Social Stress Test (TSST [10]), as well as other stress and relaxation activities including listening to functional music and a progressive muscle relaxation task. We show that earbud PPG can be used to reliably estimate IBIs in order to extract heart rate and heart rate variability. After estimating data quality and carefully filtering physiological data, we develop an explainable stress detection model based on physiological sensor data from the earbud. We show that a random forest

¹Digital Health Lab, Samsung Research America, Mountain View, California, USA, m.rahman2@samsung.com

²Pursued this work while interning with Samsung Research America. University of Washington, Seattle, WA, USA, xuhaiXu@uw.edu

³Department of Psychiatry, University of San Francisco, USA, wendy.mendes@ucsf.edu

*Both authors contributed equally



Fig. 1: Photoplethysmography (PPG), accelerometer, and core body temperature sensing using commodity earbuds.

classifier on heart rate and heart rate variability features can detect stress with 91.30% recall, 80.69% precision, and 85.12% F1-score using leave-one-subject-out (LOSO) cross-validation. We also compare the earbud model performance with the reference data. Finally, we analyze the accuracy improvement and the relative importance of adding the respiratory features and the core body temperature features to the model. Our work presents the feasibility of leveraging earbuds as a promising form factor for longitudinally robust stress monitoring.

II. RELATED WORKS

Stress detection has been widely investigated in the area of physiological signal measurement. Heart rate variability (HRV), electrodermal activity (EDA), electromyography (EMG), respiration and somatic movement are the most widely used physiological signals for stress detection [1], [4]. Previous studies [2], [3] show that it is feasible to classify stress (stress vs non-stress) with 90% accuracy by fusing various physiologic signals, e.g., HRV and respiration. Using ECG and respiratory inductive plethysmograph sensor embedded in a chest band, Hovsepian et al., [3] have detected stress with 90% accuracy and their model outcomes showed 0.72 correlation with the self-reported stress captured in free-living conditions. Using a 14-channel head-worn EEG system, another work reported 89% accuracy for classifying four stress levels [5]. However, EEG recording device is not suitable for daily use due its weight, design, and lack of portability [6], and chest bands are not comfortable to wear for everyday use [7]. Recently, the earbud form-factor is getting popular in the research community since it has great potential to continuously monitor users health [18], [16], [15] and behaviors [19]. In this paper, we present a commodity earbud that can passively sense PPG, motion, and core body temperature (CBT) and detect minute-by-minute physiological responses to stress arousal.

III. METHODS

A. Study Design

We conducted a single-arm, within-subject lab examination with 18 healthy, young individuals (aged 19-22). We have selected the healthy, young age group to minimize variations in stress responses due to comorbidities and other demographic reasons. Each volunteer sat for a quiet baseline period and then completed various stress tasks that included mild, moderate, and intensely stressful tasks. They also

performed relaxation tasks including listening to music, both calm and exciting, as well as progressive muscle relaxation. At the beginning of the study, the participants sat for a 5 minutes resting baseline recording. The participants then performed a multiple object tracking task for 3-minutes, followed by 3-minutes of progressive muscle relaxation. We then described the speech task, which included introducing participants to evaluators who would observe the speech. We described the speech task as a speech about “their first kiss”. We then let participants mentally prepare for the speech for 3-minutes and then evaluators reentered the room and sat across from participants. The research assistant then delayed the starting of the speech task to increase stress associated with the participant starting his/her the speech. We collected 3-minutes of data during this heightened anticipation period and then told participants they did not have to complete the speech task. We then collected 3-minutes of post-task recovery, followed by 4-minutes of calm music, 4-minutes of exciting music, and then the last task was a 3-minute arithmetic task called the reverse digit span task. We collected a final 3-minutes of resting recovery. Between each task, there was at least a 2-minutes gap for the subject to return to the baseline and to provide instructions for the next task. We administered the PANAS [13] questionnaire after each task to capture perceived stress in addition to the physiological stress responses. We developed an Android app to collect the earbud sensor data and annotate the timestamp of each task on a Samsung Galaxy S10e smartphone.

We collected physiological data both from the earbuds and Biopac equipment. Earbud sensors were sampled at the following rate: PPG at 25Hz, accelerometer at 50Hz, CBT at 1Hz. Reference devices were sampled at 1000Hz, and blood pressure was obtained typically at least once during each task. Impedance cardiography used a band electrode system with two Mylar tapes encircling the neck and two around the torso. ECG was obtained using a standard Lead II configuration. We obtained ear pulse transit time and skin temperature from the distal phalange of finger four. There was a cuff on the dominant upper arm to measure blood pressure. The subject’s head, hand, and torso were relatively stable in most of the tasks to reduce motion artifacts.

B. Data Analysis and Stress Detection

Data Preprocessing: We first sorted earbud samples based on the timestamp assigned during the sampling on-device to handle the dropped packets, out of sequence packets, or the delay in bluetooth communication. We then segmented the earbud sensor data based on the start and the end timestamp of each task. We also aligned the earbud data segments with the reference data segments and verified the quality of both. Earbud signal quality could be poor when it is not worn properly in the ear, if there is a poor fit, or due to excessive head motion. Excessive body and hand motion also affect the reference PPG signal. We only analyze the segments when both the earbud and the reference data showed acceptable signal quality from visual examination.

Feature Extraction: We first extract the inter-beat-intervals

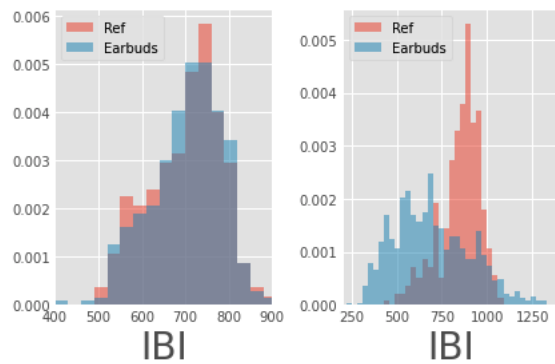


Fig. 2: Comparison between the inter-beat-interval (IBI) data from the earbuds and the reference data. X-axis shows IBI duration in milliseconds and y-axis presents normalized frequency. Left one is from the subject where the IBI distributions are identical (based on Kolmogorov-Smirnov test) and the right one is from the subject where the two distributions are different.

(IBIs) from raw earbud PPG signal using the open source Python library called heartPy [11]. We compute minute-by-minute heart rate (HR) and heart rate variability (HRV) from extracted IBIs using another library called pyHrv [12]. Heart rate variability features include both time domain features such as root means squared standard deviation (RMSSD), standard deviation of normal IBIs (SDNN), number of normal IBIs greater than 50ms (NN50), pNN50, total variation (TINN), frequency domain features including normalized low-frequency power (LF), high-frequency power (HF), LF/HF ratio, very low frequency power (VLF), and Poincaré plot features such as SD1 (standard deviation of the Poincaré plot perpendicular to the line-of-identity, describes short-term variability caused by respiratory sinus arrhythmia [22]). Previous studies show that these features are correlated with psycho-social stress [3], [12], [23]. We also extract IBIs, heart rate, and heart rate variability features from the reference signal using AcqKnowledge and Mindware software. We focused on heart rate, heart rate variability, respiratory rate, and respiratory amplitude. However, we did not include the impedance and blood pressure data given the current earbud sensors cannot extract those features.

Feature Quality Estimation: We then compare the distribution of the IBIs between the earbud and the reference data. We perform a Kolmogorov-Smirnov (KS) test to identify whether the IBI distributions were similar. If they are similar, we consider the earbud IBI data acceptable. Figure 2 shows one example of good quality IBI data (left) and poor quality IBI data (right). We observe that IBI data from four of the subjects' earbuds have significantly different IBI distributions from the reference data. We further visualize the raw PPG data from those four subjects and observe that the quality of the earbud data was poor due to the placement. These findings were consistent with previous studies on wearable physiological sensing [7].

We further compare quality of the minute-by-minute heart

rate data from the earbuds and the reference data for the subjects when the IBI distributions are acceptable. One such comparison is shown in Figure 3. It shows that the heart rate change trajectory is similar for both. Heart rate elevation during the stressors were similar and the heart rate recovery trends are also similar during relaxations. It gives us higher confidence that the earbud PPG is sensitive enough to capture autonomic nervous system response due to stress exposure. We compute Pearson's correlation between the earbud and the reference data. We observe that there were strong, positive correlation between them with average 0.70 ± 0.16 correlation coefficient. For comparison on the strength of correlation, one previous study showed that Pearson's correlation between heart rate from commonly used physiological data monitoring chest band called Zephyr HRM monitor and Biopac reference data was 0.77 [14].

Feature Selection: After the careful cleaning and screening of the physiological features, we further analyze the trends (increase or decrease) of each feature during stress and relaxation activities. Figure 4 shows the multi-modal physiological responses during stress and the relaxation task. We select the physiological features that show stronger response and in the expected direction. For example, the root mean square of differences between successive inter-beat-intervals (RMSSD) describes short-term heart rate variability. Decreased RMSSD value indicates increased stress. Ramaekers, et al., [21] emphasize that a higher expression of negative stress is related to higher vagal tone. We have selected the HRV features that show the appropriate responses.

Stress Classification: We designed the stress detection study as a binary classification problem. We assigned positive (stress) class to the minute-by-minute heart rate and heart rate variability features from the known stressors including dot-tracking, speech, and mental arithmetic task. Whereas we assigned the negative (non-stress) class to features from the baseline and relaxation tasks including calm music and resting baseline. This resulted in 473 data rows where each data row represents features extracted from one minute of continuous physiological data. Among them, 186 data rows were in positive and 287 were in negative category. Since our target is to develop a generic stress detector, we designed a leave-one-subject-out (LOSO) cross-validation experiment where we leave both the stress and non-stress data from one subject out for testing. We then train a Random Forest classifier with remaining subjects' data and then apply the model on the left-out subject's data which is completely unknown to the model. This allows us to estimate the model accuracy that might be more generalizable to future unknown subjects. To avoid over-fitting of the model, we keep the tree depth low (maximum depth as 4) and use only 100 estimators in the Random Forest model. We use the same parameters for training the model on the reference data and the earbud data for performance comparison.

IV. RESULTS AND DISCUSSIONS

Table I compares the stress detection accuracy between the earbud and the reference data model. We observe that heart

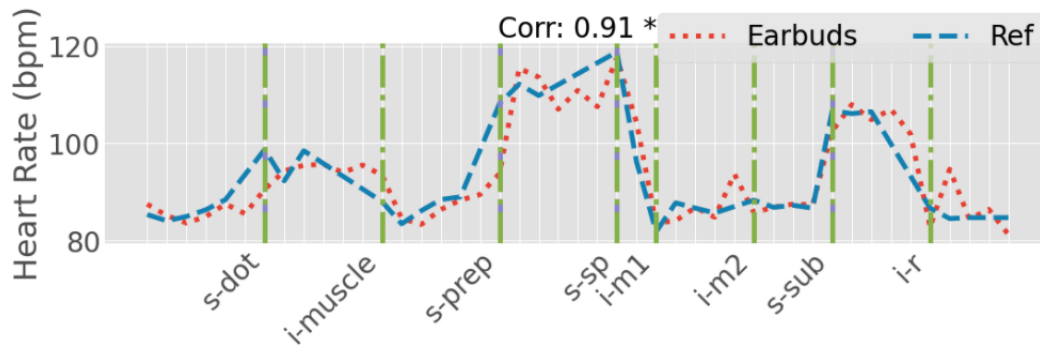


Fig. 3: Minute-by-minute heart rate comparison between the earbud and the BioPac reference for one participant. Dot tracking, speech preparation, speech delivery, serial subtractions are labeled as s-dot, s-prep, s-speech, s-sub respectively. Muscle relaxation, music1, music2, and recovery are denoted as i-muscle, i-m1, i-m2, i-r respectively. We observe that the earbud PPG based estimate is similar to the BioPac reference data. The correlation between these two heart rate time-series data is significantly positive ($\rho = 0.91$)

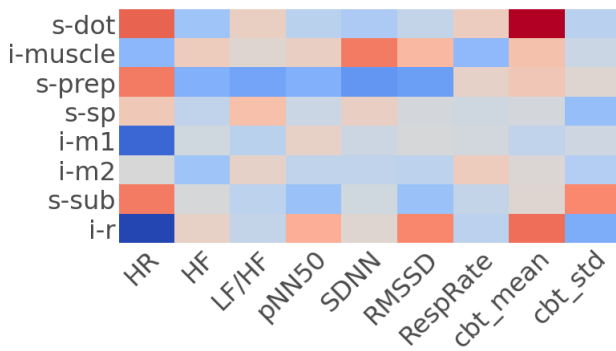


Fig. 4: Trends (increase or decrease) of the physiological responses during stressful and relaxing tasks. Red color indicates increase and blue color indicates decrease. Intensity of the color indicates the level of change.

TABLE I: Stress detection accuracy comparison between the models developed on earbud data and Biopac reference data.

Combination	Accuracy	Precision	Recall	F1-score
HR+HRV (Biopac)	86.74%	84.51%	96.25%	89.86%
HR+HRV (EarBud)	80.52%	80.69%	91.30%	85.12%

rate and heart rate variability features extracted from the reference PPG device can detect stress with 86.74% accuracy, 96.25% recall, and 84.51% precision. With the same feature set, the model trained on the earbud data shows 80.52% accuracy, 91.30% recall, and 80.69% precision. We notice that recall is comparatively higher. It means that the model is more sensitive to the stress class. It is useful because the stress events may be less frequent in general and the model should capture more events to create comprehensive insights for better stress management.

We further perform feature ablation experiments to understand how the stress detection accuracy can vary across multi-modal physiological sensors. The current earbud has PPG, accelerometer, and core-body temperature sensors.

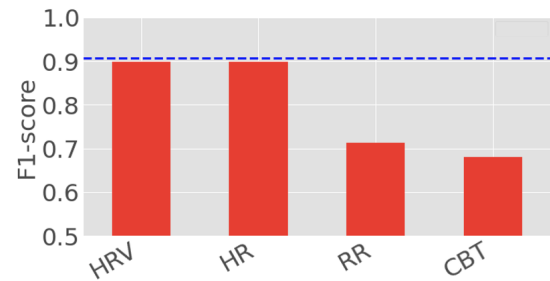


Fig. 5: Test F1-score for classification with each type of feature. HR: heart rate, HRV: heart rate variability, RR: respiratory rate, CBT: core body temperature. The blue dotted line indicates the F1-score when all features are combined together.

Several studies show that respiratory responses are useful for stress detection [3]. We have extracted respiratory rate using earbud accelerometers using our recently developed algorithms [16], [15]. Hence, we also compare the stress detection accuracy for multi-modal physiological features including heart rate, heart rate variability, respiratory rate, respiratory amplitude (depth), and core body temperature. In this study, we compute minute-by-minute mean, standard deviation, minimum and maximum of the core body temperature to fuse all three modalities together.

Figure 5 shows the findings of our feature ablation study. Using all three types of features, our model detects stress with F1-score 90.78% (precision 85.03% and recall 97.61%). We observe that heart rate and heart rate variability are the most important features for distinguishing stress followed by respiratory responses and core body temperature.

Moreover, we explore possible reasons for the accuracy differences between the earbud and the reference data, and discuss ways to further improve the accuracy. One of the major differences is the PPG sampling rate. Earbud PPG sampling rate is 25Hz and the reference PPG is sampled at 1 KHz. Battery consumption for passive, continuous PPG sampling on earbuds can be expensive. We need to find

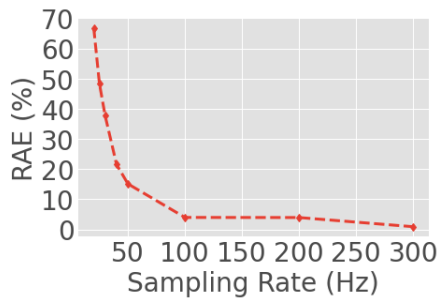


Fig. 6: Sampling rate vs relative accuracy error (RAE) for RMSSD heart rate variability feature estimation.

a trade-off between the sampling rate and the quality of measurements. To understand how the lower sampling rate could affect the quality of the measurements and inferences, we analyze HRV estimation accuracy with varying sampling rates on the reference data. We use Relative Accuracy Error (RAE) as a metric which is used in a recent study [17]. We compute the accuracy using the following equation:

$$RAE = \frac{|X_d - X_{master}|}{X_{master}} \quad (1)$$

where X_d is the HRV feature extracted after downsampling, X_{master} is the corresponding master reference data sampled at the maximum rate of 1 KHz and considered as the highest possible measurement quality.

Figure 6 shows that RMSSD measurement error becomes lower with the increase of the PPG sampling rate. Accuracy error drops well below 5% when the sampling rate is 100 Hz. We could not do this study with 100Hz due to the hardware restrictions on the commodity device. However, we will explore higher PPG sampling rate on earbuds in the future.

V. CONCLUSIONS

We demonstrate that it is feasible to detect stress arousal using multi-modal physiological sensors embedded in earbuds. We present an explainable model that detects physiological stress arousal with 85.12% F-1 score. We compare the performance with gold-standard reference data and show that earbud model performance is close to the reference model (F1-score difference is only 4.74%). Our study shows the potential of using earbud as a passive stress monitoring device that can help user better manage stress in everyday life.

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